

# HDP – Working Group Cargo Safety Concept

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Created by



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# I. Introduction

System safety considers engineering as well as management principles, criteria and techniques to consider all safety aspects throughout all phases in the systems life cycle. In addition, all the constraints in terms of cost-effectiveness, operations and timeline need to be taken into account.

Following the safety vision the first step as part of the conceptual design phase is defining the safety concept for cargoloop. The primary objective of the safety concept is to derive the functional safety requirements for the cargoloop system and its subsystems that are critical for the preliminary design phase.

In order to derive these safety requirements a set of intermediate steps are necessary such as determining the process to follow, the boundary and key assumptions of the cargoloop system, the scope of the hazard and risk analysis, identification of the conceptual solutions / design choices and selection of most suitable solutions for risk and hazard treatment within the conceptual phase.

In this document all these intermediate steps are described. In addition a set of safety requirements based on the conceptual design solutions for risk treatment are derived. The document lists a set of recommendations for future work as well.

# II. References

| # | Reference                                                                                             | Version |
|---|-------------------------------------------------------------------------------------------------------|---------|
| 1 | FAA AC 25.1309-1A: System Design Analysis Guidelines                                                  | -       |
| 2 | SAE J1739: Potential Failure Mode and Effects Analysis in Design                                      | 2002    |
| 3 | SAE ARP5580: Recommended Failure Modes and Effects Analysis Practices for Non-Automotive Applications | 2001    |
| 4 | MIL-STD-882D: Standard Practice for System Safety (Appendix A)                                        | 2000    |
| 5 | Aircraft System Safety – kritzinger (Chapter 3)                                                       | 2017    |
| 6 | IEC 61508                                                                                             | -       |
| 7 | EN 50126                                                                                              | -       |

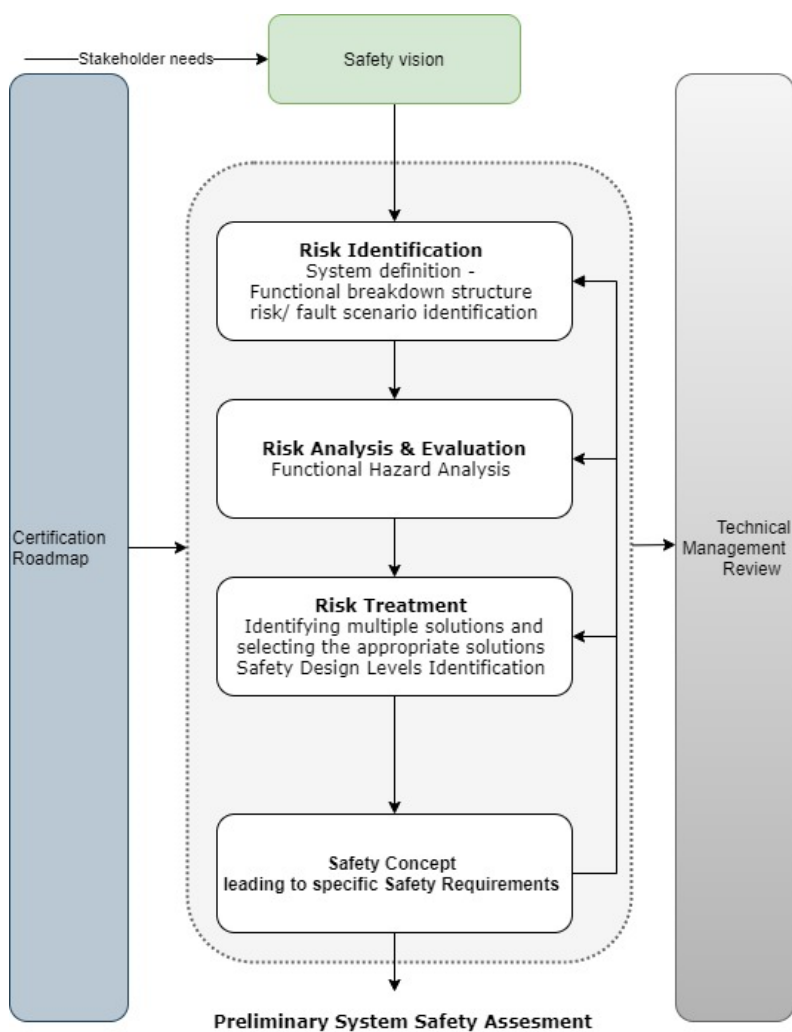
# III. Safety Concept Development Process

The safe-by design process for hyperloop follows the product development phases within its integral iterative approach. As described in the safety vision a certain set of steps will be followed during each of the product development phases.

For the safety concept development, following common well-known practices [ISO31000, SAE ARP 4761,EN50126] a set of steps have been identified:

- Safety vision
- Risk Identification
- Risk Analysis & Evaluation
- Risk Treatment
- Safety Design Level Identification
- Safety concept leading to specific requirements

The process of the safety concept development is depicted in Figure 1 below:



*Figure 1 System Safety Context*

In each step of the process several deliverables are to be created per each phase in the product development. In the Table below the deliverables and the associated review process is depicted for the conceptual design phase.

| Phase | Process Activity | Deliverable | Review |
|-------|------------------|-------------|--------|
|-------|------------------|-------------|--------|

|                         |                            |                                                                                                                                           |                                |
|-------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Conceptual Design Phase | Safety Vision              | <b>Safety Vision</b>                                                                                                                      | <b>Verification</b>            |
|                         | Risk Identification        | <b>Functional Breakdown Structure (FBS)</b><br>System/Sub-system Design Definition (SSDD-1)                                               | System Functional Review (SFR) |
|                         | Risk Analysis & Evaluation | <b>Functional Hazard Analysis</b><br>System/Sub-system Design Definition (SSDD-1)                                                         | SFR                            |
|                         | Risk Treatment             | <b>Analysis of solutions, selection of measures</b><br>Safety Design Level Identification<br>System/Sub-system Design Definition (SSDD-1) | SFR                            |
|                         | Safety Concept             | <b>Safety Concept</b><br>System/Sub-system Design Definition (SSDD-1)                                                                     | SFR                            |

Table 1

## Risk Identification

Risk identification is the first step in:

- identifying the sources of risk, system/component internal/external conditions that can cause a risk event
- classifying the impact to the system of each identified risk.

During this step it is important to consider **particular risk** scenarios due to unforeseen events (e.g. fire, natural disasters, accidents, attacks on the system) and the degree to which these scenarios must be mitigated. Such scenarios can lead to multiple functions failing and therefore need to be considered further in combination with the functional analysis of the system (Fault tree analysis). Doing so leads to a thorough understanding of the potential risks the system (may) encounters and an integral approach to safety.

## Risk analysis & evaluation

In the concept phase the functional breakdown is used as the framework to analyze risks to the system. Every function is evaluated for all conceivable failure modes during critical operation phases to analyze the failure effects. The failure effects are categorized based on severity. Furthermore, the likelihood of each failure is analyzed and categorized. Often a failure can have different causes, thus each of these is listed and estimated. Based on the combination of likelihood and severity, the criticality of the risk is obtained.

By performing the FHA (Functional Hazard Analysis) safety is integrated into the conceptual design and requirement of the system. During the later design phases (preliminary & detailed designed) an FMEA (Failure Mode Effect

Analysis) will be performed based on the systems breakdown to further evaluate the safety of the design. This way solutions and requirements can be set at every system level (from system to component).

Risk assessment matrix is to be defined (see also Section ). Furthermore, a documentation of the assumptions and acceptance criteria for the analyzed risks will be documented. Seen that the detailed design is not yet finalized at this stage the risk assessment matrix will need to be revisited in the later stages. Risk analysis & evaluation is an iterative process.

## Risk Treatment

In this step of the process multiple solutions are generated and various measures in reducing and mitigating the risk to acceptable level are identified. The most suitable measures are identified by considering the technical, operational and economically viable solutions. Safety design levels will be specified. (recommendation)

## Certification roadmap

Certification roadmap is a description of the relevant steps, applicable standards, regulations and directive to come to a certifiable hyperloop system / prototype.

In the rail sector the certification is performed by independent assessors accredited as notified bodies. In aerospace EASA can grant privileges to a manufacturers, design organization, maintenance company to perform self-assessment. For all other organization it is EASA that performs design certification.

## Technical Management Review

These are the review processes as described at each phase of the product development. Safety is an integral part of these processes.

# IV. Cargoloop System Description

The Cargoloop system primarily entails multiple coordinated (1) vehicles (2) containing various types of payload (3) that move (4) within a network of guideway systems (5), all being safe and available (6) to provide their functions at all times.

The detailed high level system breakdown can be seen in the figure below:

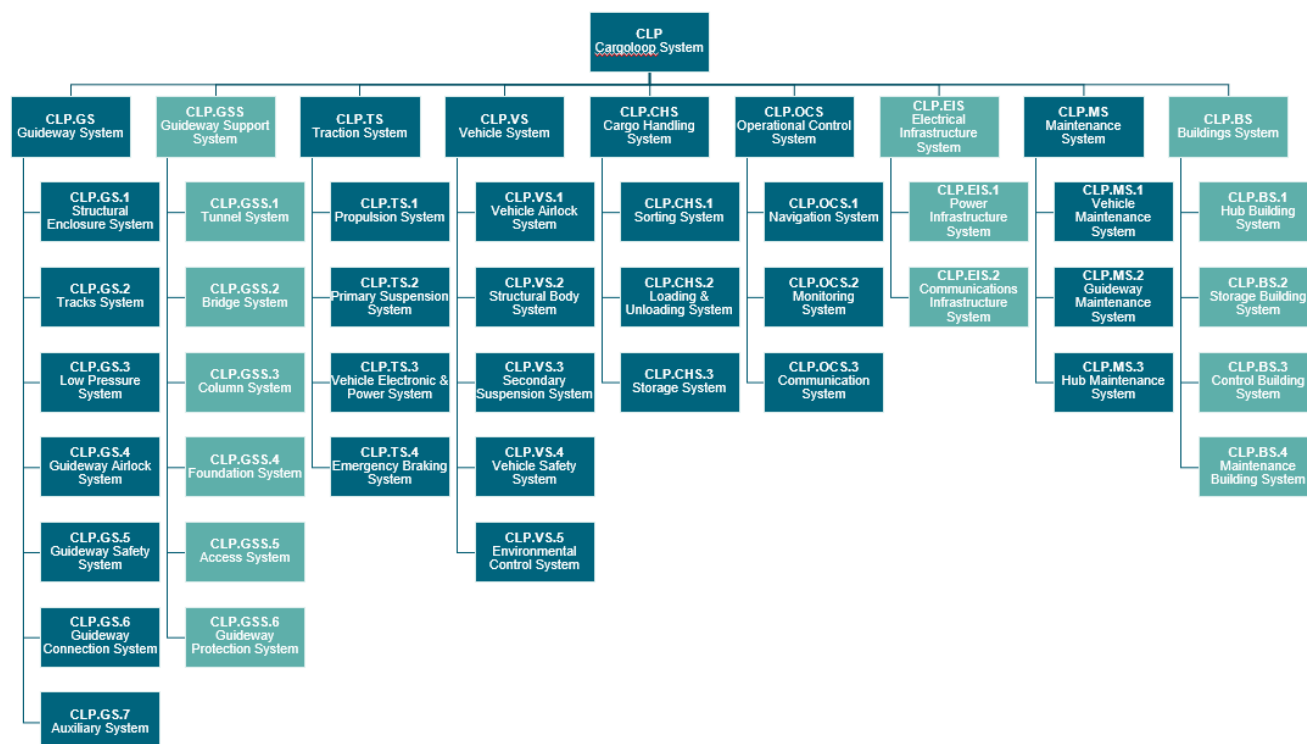


Figure 2. System Breakdown Structure

A set of concepts and possible solutions are considered out of scope. A good example is the platooning within operations.

The functional breakdown structure depicted in

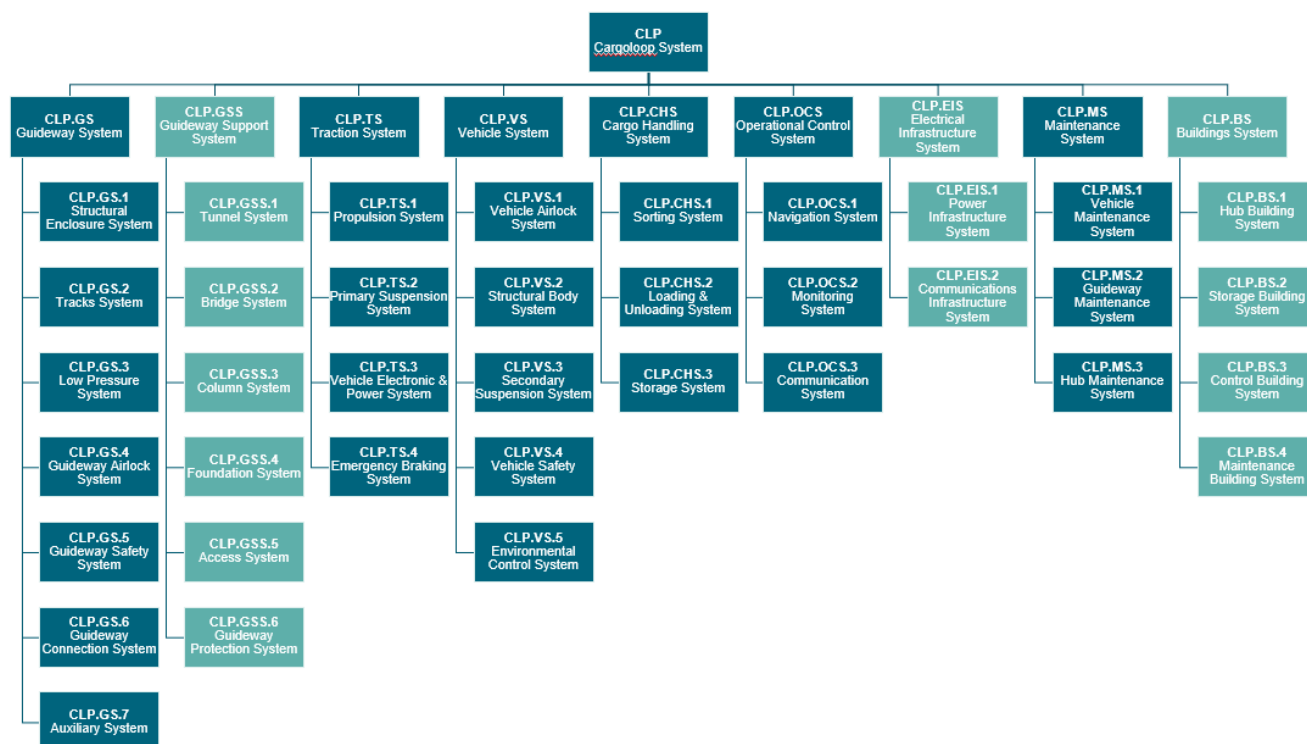


Figure 2 is used as the basis for the risk identification and risk analysis.

## V. Key Assumptions

This section provides a description of the baseline system 2.0 that is used for risk identification, risk hazard analysis and safety concept solutions in the conceptual stage.

The **generic key assumptions** on the cargoloop system covers six main systems:

- The **vehicle system (VS)** that is a container for the payload and provides all the necessary means to support the payload from the external environment.
- The **guideway system (GS)** that provides a network of enclosed tracks linking various locations, so that the vehicle can move efficiently between them.
- The **tractive system (TS)** which is the interface between the vehicle and the guideway and is the core system providing suspension and movement of the vehicle
- The **cargo handling system (CHS)** which provides all activities regarding the payload prior to the start of the vehicle journey and after the end of the journey.
- The **operational control system (OCS)** which provides the necessary coordination between multiple vehicles moving between various locations and containing various payload
- The **maintenance system (MS)** which ensures that all systems are available to provide their functions.

It should be noted that these are safety-related concept decision. With the further development and based on (safety) concept decisions the system will undergo changes. As such the recommendation is to update the baseline and key safety assumptions after the risk treatment analysis.

The cargoloop **key safety assumptions** for baseline 2.0 are:

- Severe failures or crashes are detected right after they take place (to avoid further damage)
- All systems are singular and there is no redundancy in place
- In case a vehicle drops, it is contained within the guideway
- For every 100km infrastructure, 500 vehicles operate

Further specifications based on the key assumptions (physical) (sub)systems are given in the respective sections below.

## Guideway System Assumptions

### *General systems/assumptions:*

- The guideway is a single or dual, completely enclosed tube system that is depressurized via pumps to a low pressure.
- Tracks are provided at the top/bottom and/or the sides of the tunnels to enable the vehicles to move and be positioned within the guideway (vertically and laterally).
- To enable the linking of multiple locations, multiple guideways can diverge from and merge into the 'main' line(s) and also curved guideways allow for turning.
- A (structural) support system connects the guideway enclosure to the ground (simple support or elevated)

### *Safety systems/assumptions:*

- Access points are provided for removal of the vehicle in case of damage
- Sections of the guideway can be isolated from the rest of the system in emergency?

## Vehicle System Assumptions

### *General systems/assumptions:*

- The vehicle is a pressurized vessel that contains and holds the payload as well as providing the right conditions for various payload types.
- A secondary suspension system protects the payload from vibrations and shock loads.
- Multiple environmental control systems such as temperature, pressure and humidity control are present within the vehicle.
- Doors and airlock systems are available to enable loading and unloading of the payload.
- The pitch, yaw and roll of the vehicle can be controlled

### *Safety systems/assumptions:*

Overweight payload can be identified by vehicle systems

## Traction System Assumptions

### *General systems/assumptions:*

- A primary suspension system is in place for the vertical and lateral positioning of the vehicle (EMS, EDS or Wheels – TBD). This system could be distributed or concentrated.
- An electric propulsion system is in place to propel the vehicle (on the infrastructure side)
- A braking system is present for nominal braking and bringing the vehicle to a stop.
- Electronic and power systems provide the necessary electricity for all traction functions. In addition, an electrical storage system is provided .

*Safety systems/assumptions:*

- Emergency braking system: system to bring the vehicle to a controlled stop in case of an emergency. Allows the vehicle to brake at a rate higher than the primary propulsion system.

## Cargo handling system Assumptions

*General systems/assumptions:*

- A loading and unloading system;
- Payload checking, sorting and storage systems are present;
- The cargo bay is pressurized;

*Safety systems/assumptions:*

Payload screening is performed to detect intentional hazards (e.g. explosives).

## Operational control System Assumptions

*General systems/assumptions:*

- A system is present to coordinate multiple vehicles within multiple routes based on scheduling by controlling their headway, speed, etc.
- Communication lines are present between vehicles and between vehicles and control unit(s)
- A monitoring system is continuously available to monitor the vehicle, payload and infrastructure

## Maintenance System Assumptions

*General and safety systems/assumptions:*

- Vehicles and infrastructure parts (Track sections and internals) can be removed from the guideway for repair/maintenance
- Inspections and scheduled service are provided
- Prediction systems are in place to detect component failure and wear of parts??
- There are specific locations for loitering vehicles

# VI. Concept Design Phase

## Functional Hazard Analysis

To identify the hazards faced by the systems, a functional approach is adopted in the conceptual phase, where a hazard is defined as: "Condition, event, or circumstance that could lead to or contribute to an unplanned or undesirable event." A functional approach is selected due to the system still being defined in terms of solutions however the functionalities can already be made clear. The Functional Hazard Analysis (FHA) is based upon the Functional Breakdown Structure (FBS) defined for the Cargoloop system.

Based on the functions defined in the in the third level of detail, failure modes are identified. A failure mode is defined as a manner in which a function can fail, which is distinctly different from a cause. As an example, a failure

mode of a sensor would be “provides incorrect data” and a cause would be “sensor has a faulty connection” which in turn would lead to the example failure mode. By using failure modes the FHA is again not dependent on specific solutions.

Each failure mode is then evaluated for its critical operation phases, where in some operation phases different effects can be expected. E.g. a levitation failure during parking is not as impactful as an levitation failure during cruise.

Using the combination of failure mode, operation phase, and system description/assumptions, the effect can be deducted. Thus, what are the effect of the failure mode occurring in the specific operation phase for the specified system. These end effects are then evaluated for severity and impact and can then be placed in the risk acceptability matrix (see next section).

Based on this evaluation it can be determined if a risk is acceptable. If it is not acceptable, measures are to be taken to mitigate the risk. These measures can focus on reducing likelihood or reducing severity (detectability is considered for now as part of likelihood reduction).

## Risk Assessment

The risk assessment process as well as the classifications for various failure modes has been based on safety standards from the automotive, aircraft, military and electronics industries (references 1 to 6). An example from reference 5, displays the typical safety objectives within the aerospace industry with a combination of severity and probability.

Table 3.3 Typical safety objectives

| Failure classification                                                   | No safety effect           | Minor                                                                                                          | Major                                                                                                                                                                                             | Hazardous                                                                                                                                                                                 | Catastrophic                                                                                                                           |
|--------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Allowable qualitative probability                                        | No probability requirement | Probable                                                                                                       | Remote                                                                                                                                                                                            | Extremely remote                                                                                                                                                                          | Extremely improbable                                                                                                                   |
| Qualitative probability definition                                       |                            | Failure conditions anticipated to occur one or more times during the entire operational life of each aeroplane | Failure conditions unlikely to occur to each aeroplane during its total life, but which may occur several times when considering the total operational life of a number of aeroplanes of the type | Failure conditions anticipated to occur to each aeroplane during its total life but which may occur a few times when considering the total operational life of all aeroplanes of the type | Failure conditions so unlikely that they are not anticipated to occur during the entire operational life of all aeroplanes of one type |
| Allowable quantitative probability (average probability per flight hour) | No probability requirement | In the order <sup>a</sup> of $10^{-3}$ to $10^{-5}$                                                            | In the order of $10^{-5}$ to $10^{-7}$                                                                                                                                                            | In the order of $10^{-7}$ to $10^{-9}$                                                                                                                                                    | In the order of $<10^{-9}$                                                                                                             |
| Required Functional Development Assurance Level <sup>b</sup>             | FDAL E                     | FDAL D                                                                                                         | FDAL C                                                                                                                                                                                            | FDAL B                                                                                                                                                                                    | FDAL A                                                                                                                                 |

Applying these standards to the Cargoloop systems, a grading scheme was devised using several Hardt Hyperloop experts. A failure mode is rated on severity and likelihood, where severity can be rated based on effects on either people, vehicles, infrastructure or hubs. Environment is not present due to the assumption of failures being contained to the infrastructure.

A catastrophic event is classified as a failure mode leading to death or severe injury, and/or more than 1 vehicle losses, and/or more than a week of infrastructure inoperability. A death or severe injury is catastrophic in line with other standards. The loss of material and operations is classified due to the cost associated with such events. The operational losses of such instances are the main driver.

A critical event is classified in the same manner where the outcomes are logically lower in severity (see Table 1). Thus respectively a minor injury, one or less vehicles lost, and less than a week of infrastructure downtime specifically.

Continuing for major the effects are respectively: no injury, repairable vehicle damage, and less than 24 hours of downtime/limited operating capability for the infrastructure.

Minor events are classified as events which lead to a small cost but are not requiring immediate action.

The descriptors of the likelihood categories (see Table 2) are purposefully kept interpretable (e.g. saying something shouldn't be probable to fail in 1e9 hours is not doable in the current concept design phase). Each is modelled after a standard used in aviation (add reference to the STANDARD used). See table for the descriptors.

These two ratings are combined in the risk acceptability matrix where it can be determined if a risk is acceptable or not. In the table a risk is acceptable if marked green, a risk is not acceptable and needs more mitigatory action when marked red. A risk is acceptable under certain conditions when marked by orange.

Table 1. Severity Classification

| Severity*    |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |                                                                                                                |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------|
| Effect       | Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rating | Notes                                                                                                          |
| Catastrophic | Failure mode effects safe system operation (causing death or severe injury) and/or involves noncompliance with government regulations and/or causes complete system/vehicle losses.<br><b>People:</b> death or severe injury<br><b>Vehicles:</b> more than 1 vehicle losses<br><b>Infrastructure:</b> route inoperable for more than one week<br><b>Hubs:</b> hub operations stopped for more than one week                                              | 5      | <i>with or without warning</i>                                                                                 |
| Critical     | Loss or degradation of primary function resulting in <u>major system degradation/damage</u> and/or <u>major property damage</u> . Defined as loss of a single vehicle or severe damage to a section of infrastructure or minor injury<br><b>People:</b> minor injury<br><b>Vehicles:</b> one or less than one vehicle losses<br><b>Infrastructure:</b> route inoperable for less than a week<br><b>Hubs:</b> hub operations stopped for less than a week | 4      | <i>Major degradation/damage would mean inoperable system or operable but with reduced level of performance</i> |
| Major        | Loss or degradation of secondary function resulting in <u>minor system degradation/damage</u> and/or <u>minor property damage</u> .<br><b>People:</b> no injury<br><b>Vehicles:</b> repairable vehicle damage<br><b>Infrastructure:</b> route inoperable for less than a day/halt of operations/limited capabilities in section<br><b>Hubs:</b> loading bay damaged                                                                                      | 3      |                                                                                                                |

|                                                       |                                                                                                                                                                                                                                                    |   |  |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| Minor                                                 | Does not cause property/system damage but may result in <i>system failure</i> and <i>unscheduled maintenance</i> or repair<br><b>People:</b> no injury<br><b>Vehicles/Infrastructure/Hubs:</b> repairable damage not requiring immediate attention | 2 |  |
| None                                                  | No discernible effect                                                                                                                                                                                                                              | 1 |  |
| <i>*How serious the effect of the failure mode is</i> |                                                                                                                                                                                                                                                    |   |  |

Table 1. Likelihood rating

| Likelihood* |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Occurrence  | Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Rating |
| Usual       | Failure almost inevitable                                                                                                                                                                                                                                                                                                                                                                                                                                           | 5      |
| Frequent    | Frequent failures<br><b>Vehicle:</b> Failure is anticipated to happen one or more times during the operational life of each vehicle<br><b>Infrastructure:</b> Failure is anticipated to happen one or more times during the operational life of each single route                                                                                                                                                                                                   | 4      |
| Occasional  | Occasional failures<br><b>Vehicle:</b> Failure is anticipated to happen one or more times during the operational lifetime when considering multiple vehicles. Not likely to happen when considering a single vehicle<br><b>Infrastructure:</b> Failure is anticipated to happen one or more times during the operational lifetime when considering a network/multiple routes. Not likely to happen when considering a single route                                  | 3      |
| Remote      | Relatively few failures<br><b>Vehicle:</b> Failure is not anticipated to happen to a single vehicles in its entire lifespan, but may occur a few times when considering the full operational life of all the vehicles in the system<br><b>Infrastructure:</b> Failure is not anticipated to happen to a single route (<250 km) in its entire lifespan, but may occur a few times when considering the full operational life of all the infrastructure in the system | 2      |

|                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                      |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Improbable                                                                                                                                                                                                           | <p>Failure is very unlikely</p> <p><b>Vehicle:</b> Failure not anticipated to happen during the entire operational life of all vehicles</p> <p><b>Infrastructure:</b> Failure not anticipated to happen during the entire operational life of all infrastructure</p> | 1 |
| <p><i>* Estimated frequency or cumulative number of failures that will occur <b>for a given cause</b> over the "intended life of the design"</i></p> <p><i>** 24 hr/day operation and single vehicle assumed</i></p> |                                                                                                                                                                                                                                                                      |   |

Table 3. Risk matrix

| Risk acceptance criteria                                                                                                                                                                                                                                                       |             |               |              |              |              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------------|--------------|--------------|
| Likelihood                                                                                                                                                                                                                                                                     | Severity    |               |              |              |              |
|                                                                                                                                                                                                                                                                                | None        | Minor         | Major        | Critical     | Catastrophic |
| Usual                                                                                                                                                                                                                                                                          | Tolerable*  | Unacceptable* | Unacceptable | Unacceptable | Unacceptable |
| Frequent                                                                                                                                                                                                                                                                       | Acceptable* | Tolerable     | Unacceptable | Unacceptable | Unacceptable |
| Occasional                                                                                                                                                                                                                                                                     | Acceptable  | Tolerable     | Tolerable    | Unacceptable | Unacceptable |
| Remote                                                                                                                                                                                                                                                                         | Acceptable  | Acceptable    | Acceptable   | Tolerable    | Unacceptable |
| Improbable                                                                                                                                                                                                                                                                     | Acceptable  | Acceptable    | Acceptable   | Acceptable   | Tolerable    |
|                                                                                                                                                                                                                                                                                | None        | Minor         | Major        | Critical     | Catastrophic |
| <p><i>*Acceptable : necessary to maintain that risk remains at this level</i></p> <p><i>*Tolerable : Only if risk reduction is impractical or if its cost is grossly disproportionate to the improvement gained</i></p> <p><i>*Unacceptable : Risk cannot be justified</i></p> |             |               |              |              |              |

## Hazard Topics

Based on the analysis of the failure modes in the FHA a set of hazard topics has been identified. Each of these hazard topics contains a number of the related failure modes.

The identified hazard topics are listed below:

1. Environmental Conditions
2. Intentional Harm
3. Tracks and Pathways
4. Guideway Structural Integrity
5. Pressure Interface failure
6. Levitation and Guidance

7. Guideway Conditioning
8. Propulsion
9. Securing Payload
10. Communication Systems Failure
11. Coordination of Operations Failure
12. Routing Failure
13. Cargo Screening/Hub Security related to Payload

## VII. Safety concept solutions

Each of the failure modes within the hazard topic has been analyzed and risk treatment measures have been identified. Solution analysis and the requirements derived has been done according to a defined template developed for this purpose. One example of this template is depicted in the Appendix.

In the following Sections each of the hazard topics is described as well as the selected solutions for risk treatment and the identified requirements.

### Environmental Conditions

This hazard group covers failure modes / environmental hazards that effect the safe operation as well as the lifetime of the systems. A combination of multiple solutions are suggested to overcome the hazards (either reduce the likelihood of them effecting operations or resulting in damages or reducing the severity of the possible damages). They are listed below:

- 1- Lightning and shock protection systems
- 2- Making the substructure earthquake-proof with different design limits for operation and damage
- 3- Detection of ice accretion on infrastructure, particularly around vents and valves
- 4- Protection of systems against animals (rodents), fungi and plants
- 5- Consideration of wind loadcases in design of the guideway and guideway support system with different design limits for operation and damage
- 6- Design against UV light and radiation for exposed systems/components
- 7- Fire protection systems within and around the guideway as well as in hubs

The requirements based on the safety concept work are given as follows:

| # | Description                                                                                                                                                                                                                                                                       | Impacted system | Impacted sub-system |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------|
| 1 | <p>A lightning protection system shall be present to:</p> <p>1-reduce the likelihood of lightning striking the guideway</p> <p>2-prevent critical structural damage to the guideway causing loss of primary function</p> <p>3-protect all electrical systems from overvoltage</p> | Cargoloop       |                     |

|           |                                                                                                                                                                                                                                                                                                                           |                                          |                                            |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------|
| <b>2</b>  | All systems shall be designed to prevent electric shock harm to personnel and systems                                                                                                                                                                                                                                     | Cargoloop                                |                                            |
| <b>3</b>  | Systems shall be designed to prevent injury and major damage to the infrastructure and vehicles from earthquakes with a magnitude of X on the Richter scale taking place for a maximum duration of Y seconds (in earthquake zones where the probability of hitting the specified magnitude is more than Z% within a year) | Guideway Support System                  | Guideway Support Structure                 |
| <b>4</b>  | Systems shall be designed not to limit the operational capability when earthquakes with a magnitude of XX on the Richter scale take place for a maximum duration of YY seconds (in earthquake zones where the probability of hitting the specified magnitude is more than ZZ% within a year)                              | Guideway Support System                  | Guideway Support Structure                 |
| <b>5</b>  | Live organisms such as animals and fungi shall not cause any major damage to vehicles, infrastructure and hubs during the full lifetime of each system                                                                                                                                                                    | Cargoloop                                |                                            |
| <b>6</b>  | Scheduled maintenance shall include inspecting for ice formation on and damage to sensitive structures as well as repair of any damage due to ice/water (e.g. corrosion)                                                                                                                                                  | Guideway system, Guideway support system | Structural enclosure system, tracks system |
| <b>7</b>  | Freezing weather conditions based on expected operating temperatures shall not result in blockage of and critical damage to the vacuum infrastructure (e.g. valves and vents)                                                                                                                                             | Guideway system                          | Low-pressure system                        |
| <b>8</b>  | Fire protection and suppression systems shall be present within and around the guideway and at the hubs                                                                                                                                                                                                                   | Cargoloop                                |                                            |
| <b>9</b>  | All systems shall be able to operate with full functionality in ambient temperatures between X to Y degC (operational temperatures)                                                                                                                                                                                       | Cargoloop                                |                                            |
| <b>10</b> | There shall be no damage to any part of the system in ambient temperatures between X and Y degC (limit temperature)                                                                                                                                                                                                       | Cargoloop                                |                                            |
| <b>11</b> | Corrosion shall be detected before it can cause damages and failure that limit the functionality of systems                                                                                                                                                                                                               | Guideway System, Guideway Support System |                                            |

|           |                                                                                                                                                                                               |                                          |  |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--|
| <b>12</b> | Winds with an average speed of X and maximum short time (Y seconds) gusts of speed Z shall not limit the operational capability of the system                                                 | Guideway System, Guideway Support System |  |
| <b>13</b> | Winds with an average speed of XX and maximum short time (YY seconds) gusts of speed ZZ shall not result in damage to the infrastructure resulting in an inoperable route for more than a day | Guideway System, Guideway Support System |  |

## Intentional Harm

This hazard group covers, risks associated with external threats and failure to control access to the system. The solutions in this category involve mostly actions to reduce the likelihood of damages or interruptions happening due to these hazards. The solutions are as follows:

- 1- Unauthorized access prevention to restricted areas
- 2- Prevent vehicles hitting the infrastructure
- 3- Shut-down or limiting the operational envelope of sections of the system in case of unauthorized access with risk of harm to the system
- 4- Multiple communication layers with firewalls (see further the communications system failure section)

The following requirements are proposed based on the suggested solutions:

| #        | Description                                                                                                                                                                            | Impacted system                          | Impacted sub-system |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------|
| <b>1</b> | Unauthorized access to operational areas (against reasonably expected threats) shall be prevented                                                                                      | Cargoloop                                |                     |
| <b>2</b> | Vehicular access to the guideway by anyone (excl. Personnel) with vehicles commonly accessible to the public shall be prevented                                                        | Guideway Support System                  |                     |
| <b>3</b> | Back-up access points shall be available for personnel to access the guideway in case of emergency                                                                                     | Guideway Support System, Guideway System |                     |
| <b>4</b> | Safety areas for storage of stationary vehicles and maintenance tools as well as performing maintenance activities shall be envisioned at hubs as well as every x km along every route | Cargoloop                                |                     |
| <b>5</b> | Damages due to intentional harm shall not be anticipated to happen to a single route (<250km) in its entire lifespan                                                                   | Cargoloop                                |                     |
| <b>6</b> | In the event of unauthorized access (with risk of intentional harm) in and around the guideway, operations shall be halted on that route                                               | Operation Control System                 |                     |

|           |                                                                                                                                                                                                    |                          |  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--|
| <b>7</b>  | In the event of unauthorized access (with risk of intentional harm) in and around the guideway, operations on adjacent routes shall enter an emergency state (with lower speed and throughput TBD) | Operation Control System |  |
| <b>8</b>  | The operational control system shall be designed to reduce the likelihood of hacking to less than x times during the operational lifetime of multiple vehicles                                     | Operation Control System |  |
| <b>9</b>  | The Operational control system shall be designed to detect unauthorized cyber access                                                                                                               | Operation Control System |  |
| <b>10</b> | The Operational control system shall be designed to shut-down or halt operations in case unauthorized cyber access is detected                                                                     | Operation Control System |  |

## Tracks and Pathways

This hazards covers risks associated with the pathway of the vehicle being compromised. This can mainly occur due to the tracks being out of tolerance (e.g. deformed, fractured, not properly attached). The tracks enable levitation and guidance, and might enable EBS/positional safety systems. Effects include collisions with parts of the infrastructure and losses of multiple vehicles due to headway being lower than the brick-wall assumption. Less severe effects include inability to EBS in a section or lack of a positional safety system in a section (only becomes relevant if primary functionality fails). Suggested solutions are listed below:

1. Prevent tracks from moving into the vehicle keep-out zones by having reliable connections (e.g. the track being prevented to move inwards by the mounting structure, redundant critical connections, safety factors on design, etc)
2. Detectability of loss of tolerances (e.g. vehicle side sensors, infrastructure inspection).
3. Mitigation of severity using vehicle positional safety system: Levitation&Guidance

| #        | Description                                                                                                                        | Impacted System | Impacted Sub-system                 |
|----------|------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------|
| <b>1</b> | The failure of structures and the tracks shall not obstruct the pathway of the vehicle when considering the lifetime of a network. |                 | Guideway System                     |
| <b>2</b> | The structures and the tracks shall not be expected to fail when considering the lifetime of a network.                            |                 |                                     |
| <b>3</b> | Loss of tolerances on the infrastructure shall be detectable before compromising functionality.                                    |                 | Guideway System, Maintenance System |

|   |                                                                                                                                                                                                                                        |  |                                                   |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------|
|   |                                                                                                                                                                                                                                        |  |                                                   |
| 4 | If a vehicle loses primary positional control, the damage shall be contained to within the guideway.                                                                                                                                   |  | Guideway System                                   |
| 5 | If a vehicle loses levitation and/or guidance functionality control anywhere in the guideway, the potential damages shall be repairable within a week.                                                                                 |  | Guideway System, Vehicle Positional Safety System |
| 6 | The guideway tracks shall not be expected to fail under nominal operation when considering the lifetime of a network.                                                                                                                  |  | Guideway System                                   |
| 7 | The guideway tracks shall be able to withstand a expectable impact from a vehicle at least once (the track breaking/needing repair is allowed) and be able to provide functionality for the vehicles behind which cannot halt in time. |  | Guideway System                                   |

## Guideway Structural Integrity

This hazard group covers risks of damage to or failure of the structure of the guideway and guideway support system. The guideway and systems supporting the guideway could lose their structural integrity which could be in a static/dynamic manner as well as effecting specific structural characteristics (deformation/stress/...). The most severe effects happen during *Drive* but in general they are not limited to a specific operating condition. In a worst case, some of the failure modes could lead to rupture of guideway sections, sudden depressurization and damage to (multiple) vehicles at the same time. The solution is a combination of sub-solutions which are listed below. They are mainly focused on proper design approaches as well as prevention and early detection.

- 1- Proper structural design based on selected design code (Full loadcase identification, safety factors on loads and material data, Safety margins on buckling and frequencies, etc.)
- 2- Periodic Inspection of structure for crack initiation, structure thinning, connector pre-tensioning and coating damage
- 3- Detection of structural vibrations, structural deformations and foundation sagging to halt operations or limit operation envelope if necessary (using sensors on vehicles)
- 4- Redundant and back-up load paths incl. for connectors (excl. guideway supports especially for elevated tubes)

The following requirements are suggested for the next phases:

| # | Description                                                                             | Impacted system                                             | Impacted sub-system |
|---|-----------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------|
| 1 | Safety factors shall be taken into account for load and material data uncertainty (TBD) | Guideway System,<br>Guideway Support System, Vehicle System |                     |

|          |                                                                                                                                                                                                           |                                                             |  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--|
| <b>2</b> | A safety margin of x Hz <u>should</u> be considered to prevent excitation of eigenfrequencies                                                                                                             | Guideway System,<br>Guideway Support System, Vehicle System |  |
| <b>3</b> | Failures of the guideway and guideway support structures <u>should</u> not obstruct the operations                                                                                                        | Guideway System,<br>Guideway Support System                 |  |
| <b>4</b> | Scheduled maintenance (intervals TBD) shall be in place to inspect and maintain the guideway and guideway support structural elements                                                                     | Guideway System,<br>Guideway Support System                 |  |
| <b>5</b> | It shall be possible to detect excessive structural deformation of the guideway tube and tracks before they lead to any failures                                                                          | Guideway System,<br>Guideway Support System                 |  |
| <b>6</b> | It <u>should</u> be possible to detect excessive and unintended vibrations of the guideway and guideway support structural elements before they result in instability and major system or property damage | Guideway System,<br>Guideway Support System                 |  |
| <b>7</b> | Structural elements within the guideway support system of a single route (<250 km) shall not be anticipated to fail during their entire operational life                                                  | Guideway System,<br>Guideway Support System                 |  |
| <b>8</b> | The Dynamic Amplification Factor (DAF) of the guideway structural elements during all operational phases and maximum throughput shall not exceed X                                                        | Guideway System,<br>Guideway Support System                 |  |
| <b>9</b> | It shall be possible to detect all signs of possible structural failures (such as cracks and thinning) before failure happens and functionality is lost                                                   | Guideway System,<br>Guideway Support System                 |  |

## Pressure Interface failure

This hazard group covers failures related to the high pressure differential during loading and unloading:

- Personnel & items can get sucked into the guideway, stresses /damages vehicle and loading platform
- Potential loss of life, vehicle and loading bay.

In order to prevent these failure it is necessary to:

1. minimize access of the personnel and unwanted objects to the cargoloop system loading/unloading area;
2. recognize this event and the cargoloop system can react accordingly

In step 1 solutions such as no walk zones with a fences (likelihood reduction) and safety belts (likelihood & severity reduction) need to be defined in the next preliminary design phase. This should be part of the workplace safety procedures.

In step 2 several solutions have been identified that allow either severity reduction for human harm, or allow faster (partial) recovery of the system by being able to close off the section that is damaged. The possible solutions are:

- Decoupling of loading/unloading from the cargoloop system (removable cargobay);
- Double pressure airlock, which is inherent to the above solution(removable cargobay);
- Rapidly opening and closing passive doors (valves);
- Air pressure vents are identified as a solution that would enable opening vents only in case of such event. This would enable depressurizing the affected sections. This is also a good solution also for isolating sections for maintenance.

The necessary safety measures in Step 1 around the cargoloop system are seen as a necessary step. In step 2 the removable cargobay is selected as it allows for likelihood and severity reduction of such hazard.

The following requirements are identified based on the selected solution:

| # | Requirement Description                                                                                                                | Impacted System (FBS) | Impacted Sub-system        |
|---|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------|
| 1 | Safety measures (e.g. physical object measures, no walk zones) shall ensure safe access to the cargoloop system loading/unloading area | Cargoloop             | Hub design                 |
| 2 | The loading /unloading of the payload shall be designed to prevent humans and/or object being drawn into the guideway                  |                       |                            |
| 3 | The unwanted objects within the guideway during loading / unloading of the payload shall be detected                                   | Cargoloop             | Infrastructure sub-systems |

## Levitation and Guidance

The levitation and guidance hazards covers failure modes which result in loss of lateral and vertical positioning of the vehicle. Due to such a failure mode several end effects are encountered: loss or damage of infrastructure and loss of multiple vehicles, indirectly also perception of safety of Hyperloop technology is tainted.

At this point in time the hazard is labelled for further investigation due to the complexity of the issue. Some important aspects are the designed for reliability of vehicles and the ability to guarantee this reliability, the allowed damages to the infrastructure due to such an event, and the allowed loss of public faith in the Hyperloop for passenger implementation.

Several solutions are possible which focus on different strategies:

1. Preventing failure to occur: system is designed in such a way failure of these functionalities is never expected. The risk lies in the ability to design and guarantee this reliability

2. Preventing failure to occur & limiting severity of damages: see 1 with the addition that damages are limited to a shorter section of infrastructure/only non-critical tracks and only the failing vehicle
3. Preventing failure to occur & not incurring damages when failure occurs: very expensive implementation (basically having two primary systems)

A requirements draft for solution 2 is presented since it is the recommended solution:

|   | Description                                                                                                                                                   | Impacted System (SBS)             | Impacted Sub-system                                                                  |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------|
| 1 | The primary positional control functionality shall not be expected to fail when considering the lifetime of a single vehicle (<1% probability).               |                                   | Primary suspension, secondary suspension, vehicle structural system, guideway system |
| 2 | The primary levitation & guidance functionality shall not be expected to fail when considering the lifetime of X vehicles (<1% probability).                  | Related hazard: Tracks & Pathways | Primary suspension, secondary suspension, vehicle structural system, guideway system |
| 3 | If the primary positional control functionality fails, operational shall not be halted for more than a week.                                                  |                                   |                                                                                      |
| 4 | If the primary suspension functionality fails, the payload bay shall remain intact & be subjected to loads low enough to demonstrate passenger survivability. |                                   |                                                                                      |
| 4 | If the primary suspension functionality fails, no large parts of the vehicle (>1 kg or >10 cm) shall come lose the vehicle.                                   |                                   |                                                                                      |
| 5 | If the primary positional control functionality fails, damage shall be limited to the guideway and not impact the environment around the guideway             |                                   |                                                                                      |
| 6 | If the primary positional control functionality on a vehicle fails, no other/less than x vehicles will be damaged.                                            |                                   | EBS                                                                                  |

## Guideway Conditioning

Guideway conditioning hazards are related to the high pressure difference present between the system and ambient conditions. Hazards exist to personnel, infrastructure and vehicles due to this. Personnel can be endangered by coming in contact with vacuum environments or high pressure gradients/changes. Vehicles and infrastructure can be damaged by (rapid) changes in pressure conditions. Solution are proposed for the different problems:

1. Vents and pumps connected to the vacuum environment should not be accessible to personnel when connected to the vacuum environment. Vents should be placed enclosed, physically shielding them. The pump needs to be physically disconnected (e.g. valves) from the vacuum environment when accessing the pumping room. The room should also be kept open to ventilation
2. Pump systems are redundant due to them being independent from each other (should have some overcapacity).
3. Valves are placed to provide measures to alleviate pressure when necessary. Redundant sensing is placed and also used to detect leaks early.
4. *(Pressure relief valves can be used in case of a rapid depressurization to limit propagation of a shockwave to limit further damage.)*
5. Measures should be place around switches to prevent a route depressurizing impacting the greater network (maybe also redundantly executed). E.g. gate valves around switches.

Requirements are given below:

| # | Description                                                                                                                                                                                                       | Impacted System | Impacted Sub-system                                                           |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------|
| 1 | When personnel is present near vacuum systems, the personnel is to be protected from harmful pressure differences caused by the vacuum system.                                                                    |                 | Vacuum system,<br>Maintenance system                                          |
| 2 | When a single vacuum pump installation fails, the vacuum system shall remain operational and the guideway shall remain at nominal pressure.                                                                       |                 | Vacuum system                                                                 |
| 3 | If the vacuum vessel of is compromised to require halt of operations in a route (e.g. pressure not maintainable within safe bounds), the remaining network shall be operational within 24 hours.                  |                 | Guideway System                                                               |
| 4 | The vacuum vessel is not expected to be compromised to the point of halting operations in a route when considering the lifetime of a single route, but may occur several times when considering the full network. |                 |                                                                               |
| 5 | When a sudden repressurization occurs (section to ambient pressure within seconds), the infrastructure system shall not incur damage to other guideway sections due to the sudden pressure rise                   |                 | Guideway System<br>(e.g. repressurizing ahead of shockwave),<br>Vacuum System |

|   |                                                                                                                                     |  |  |
|---|-------------------------------------------------------------------------------------------------------------------------------------|--|--|
| 6 | A vacuum installation should not fail more than a couple of times when considering multiple installations over their full lifetime. |  |  |
| 7 | When a sudden repressurization occurs, vehicles shall incur limited (e.g. only to covers) & repairable damage                       |  |  |
| 8 | A sudden repressurization of the vacuum vessel is not expected to occur when considering the lifetime of a network.                 |  |  |

## Propulsion & Braking

| # | Description                                                                                                                                                        | Impacted system        | Impacted sub-system |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------|
| 1 | The propulsion system shall be replaceable within x days for a section of y meters                                                                                 | Propulsion system      |                     |
| 2 | The propulsion system should not be expected to fail more than Y times when considering the lifetime of X vehicles (<1% probability) or z length of infrastructure | Propulsion system      |                     |
| 3 | The EBS system shall prevent collisions with vehicles ahead which have lost active control (No brick wall assumption)                                              | Propulsion system, EBS |                     |
| 4 | If the primary propulsion fails full functional operations shall be reinstated within Xh                                                                           | Propulsion system      |                     |
| 5 | Redundant power systems shall avoid complete power failure                                                                                                         | Propulsion system      | Electrical system   |
| 6 | (If the levitation & guidance functionality fails, the vehicle should brake at a high rate (~1g +) )                                                               | EBS                    |                     |

## Securing Payload

Securing payload hazards group are failure modes related to securing payload in all three space dimensions within the vehicle (x,y,z direction). Failing to secure payload leads to damage to vehicle, sudden forces in the vehicle resulting in the worst case in damage to infrastructure, temporary stop of operations. The solution identified for this hazard is in securing the system within the vehicle such that:

- Cargo is secured in x,y,z within the cargobay

- A 'pallet shaped' slide (also in height) in cargo bay is seen as a suitable measure;
- The empty space should be filled

The identified requirements are listed below:

| # | Description                                                                                                                          | Impacted System (SBS) | Impacted Sub-system |
|---|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------|
| 1 | The cargo shall be secured within the cargobay                                                                                       | Vehicle               | Loading/Unloading   |
| 2 | The movement of cargo in any x,y,z position within the vehicle shall be prevented during the entire trip                             | Vehicle               | Loading/Unloading   |
| 3 | If the cargo gets loose (during loading/unloading or during the trip) it shall not cause damage to the vehicle and/or infrastructure | Loading/Unloading     |                     |

## Coordination of Operations Failure

This hazards group contains the failure modes related to the failure of the coordination of the state of the vehicles such as: the vehicle fails to take into account vehicles ahead / behind in the same guideway, vehicles are too fast near the hub or the switch, coordination fails to take into account all the vehicles, vehicle does not execute co-ordination commands.

Several solutions have been discussed:

1. Vehicle deploys autonomously EBS in a failure event;
2. OCS to Guideway signaling designed for safety
  - a. Operating System dictates the signaling system part of the infrastructure
  - b. OCS is connected to Guideway System through fixed connection
    - i. Signaling system is designed for safety:
  - c. Signaling is designed for safety and implemented into key locations in the guideway
    - Various possibilities in terms of signs have been discussed (e.g speed signs for a recommended speed range, barcodes, closed loop signaling systems (such as e.g. traffic lights))
  - d. Sensor on the vehicle (wireless) to read/interpret the signaling;
  - a. Vehicle should have additional object recognition on board to detect the objects / vehicles in front of its trajectory;
3. Operating Control System(OCS) is able to communicate with all relevant systems and subsystems in the cargo loop through primary robust communications infrastructure .
  - i. This infrastructure will enable all information flows between the OCS, guideway and vehicles not only for the safety, but also for example for maintenance;
4. Additional communication systems to enable vehicle to vehicle communications will enable a hyperconnected cargoloop system which would be beneficial for safety, operations and maintenance.

Having the vehicle to perform emergency brake at all times is seen as a solution for this hazard. However, in addition a robust reliable solution for coordinating the state of the vehicles still needs to be determined. The solution to the failures of the coordination of the state of the vehicles requires the implementation of a robust and highly reliable communication systems infrastructure. This infrastructure will be the key enabler for the coordination of the state of the vehicles functionality of the Operating Control System (OCS) during the entire lifetime of the system. It must be noted that the usability of the radio / wireless communications systems in an hyperloop environment need to be evaluated and proven to work in such an environment. This is due to the fact that these systems have not been designed for a hyperloop application. Some hyperloop developers have published the first results<sup>1</sup> in this field. Specific communications systems are also designed for railways<sup>2</sup> and aviation. Within this solutions space a layered approach could be considered. Starting from a simple solution and scaling up to a fully connected hyperloop. This would entail:

- OCS communicates to the guideway and there is a fixed predefined basic signaling to the vehicle, which the vehicle is able to identify;
- Coordination of the state of the vehicles through primary robust communications infrastructure between the OCS, guideway and vehicles
- Fully connected hyperloop with an extensive and robust communications infrastructure that allows for vehicle to vehicle communications;

The identified requirements are listed below:

| # | Description                                                                                                                      | Impacted System (SBS)                                                   | Impacted Sub-system                                                            |
|---|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1 | The OCS shall be able to coordinate the state of the vehicles during the entire lifetime of the system                           | Operation and Control System                                            | EBS, Vehicle and Guideway Safety,<br>Communication System<br>Navigation System |
| 2 | A full failure (all coordination capability in a system) of the OCS is not expected when considering the lifetime of the system. | Operation and Control System                                            | EBS, Vehicle and Guideway Safety,<br>Communication System<br>Navigation System |
| 3 | A full failure (all coordination capability in a system) of the coordination system shall lead to a safe halt of operations      | Operation and Control System, Traction System, Guideway System, Vehicle | Communication System<br>Navigation System<br>EBS, Vehicle and Guideway Safety, |

<sup>1</sup> <https://www.mdpi.com/1996-1073/14/4/983>

<sup>2</sup>

|   |                                                                                                                                                                                                                                                                                                                                   |                                                                         |                                                                                |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 4 | A local failure (a set of the system, e.g. x vehicles, y sections of infrastructure, of coordination capability) of the Operational Control System (vehicle or infrastructure side) is not expected when considering the lifetime of a route or a set of vehicles (~hundreds of vehicles).                                        | Operation and Control System, Traction System, Guideway System, Vehicle | Communication System<br>Navigation System<br>EBS, Vehicle and Guideway Safety, |
| 5 | A local failure (a set of the system, e.g. x vehicles, y sections of infrastructure, coordination capability) of the coordination system should lead to a safe (no collision), (reduced capability) continuation of operation (e.g. sectioned infrastructure with redundant coverage, autonomous operation when not coordinated). | Operation and Control System, Traction System, Guideway System, Vehicle | Communication System<br>Navigation System<br>EBS, Vehicle and Guideway Safety, |
| 6 | The coordination system shall not expect to cause a vehicle to infrastructure collision when considering the lifetime of the system.                                                                                                                                                                                              | Operation and Control System, Traction System, Guideway System, Vehicle | Communication System<br>Navigation System<br>EBS, Vehicle and Guideway Safety, |
| 7 | The OCS shall not expect to cause a vehicle to vehicle collision when considering the lifetime of the system.                                                                                                                                                                                                                     | Operation and Control System, Traction System, Guideway System, Vehicle | Communication System<br>Navigation System<br>EBS, Vehicle and Guideway Safety, |
| 8 | If a hazard or failure occurs, the coordination system shall limit further impact by coordinating emergency procedures (e.g. surrounding vehicles, EBS, infrastructure safety measures, etc).                                                                                                                                     | Operation and Control System, Traction System, Guideway System, Vehicle | Communication System<br>Navigation System<br>EBS, Vehicle and Guideway Safety, |
| 9 | If the coordination system fails (e.g. communication failure, coordination center failure), vehicles & infrastructure can autonomously determine action to safely halt/limit/slow down/continue operations.                                                                                                                       | Operation and Control System, Traction System, Guideway System, Vehicle | Communication System<br>Navigation System<br>EBS, Vehicle and Guideway Safety, |

|    |                                                                                                                                                                                                                                                   |                                                                         |                                                                                |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 10 | The coordination system shall be designed to prevent malicious interference to a degree it would not be expected to halt the systems operations more than X times during the lifetime of the system (cyberattacks, tampering with communications) | Operation and Control System, Traction System, Guideway System, Vehicle | Communication System<br>Navigation System<br>EBS, Vehicle and Guideway Safety, |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------|

## Routing Failure

This hazard group entails features related to the wrong routing of the vehicles into guideways. This can either happen during the Drive or during loading / unloading. Several solutions have been identified:

1. Hub design: roundabout design unidirectional flows only solving the wrong guideway
2. Program vehicle to follow a sequence of switches on a given route
3. Flow control /navigation feature real-time

Implement flow control /navigation feature real-time from the operating system to the vehicles to enable rerouting in the event of failure to routing. The traffic flow management system shall take into account the network layout and control the flow of the vehicles within the guideway accordingly. The solution assumes that the communication infrastructure is available and therefore there is a possibility to command the vehicles accordingly. A redundant system is necessary to bring the traffic flows to a safe stop. The flow control through a navigation system is seen as the solution. This system could be potentially used for coordinating the state of the vehicles. The identified requirements are listed below:

| # | Description                                                                                                                                                                                                                 | Impacted System (FBS)     | Impacted Sub-system                                |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------|
| 1 | A routing failure caused by the coordination system is not expected to happen when considering the lifetime of the system.                                                                                                  |                           |                                                    |
| 2 | A vehicle that is wrongly commanded into the wrong guideway shall be detected by the coordination system and the vehicle                                                                                                    | Cargoloop: TS, VS and OCS | Operation and Control System, Communication system |
| 3 | If a vehicle is taking a wrong guideway (e.g. due to malfunctioning, intentional harm), the operating and control system shall be able to alter and/or halt the flow of the traffic / operations in that particular section | Cargoloop: TS, VS and OCS | Operation and Control System, Communication system |

## Communication Systems Failure

This group covers the failure of the communications system. Communication system is a key enabler of the operations and control system. Failure of the communications shall lead to the temporary stop of operations, could potentially lead to loss of vehicles, damage to infrastructure.

It must be noted that the usability of the communications systems in an hyperloop environment need to be proven to work in an hyperloop environment. This is due to the fact that these systems have not been designed for a hyperloop application.

Several potential solutions are identified:

1. Vehicle stops utilizing the EBS
2. Signaling system

These signs could be for example hardcoded in the tracks such that the vehicle has some basic safety guidance in case of communications failure. Such signs could be:

- a. slow down by 75%,
- b. go to next safety parking, etc.

The rules should be standardized and known.

3. Fail-safe system redundant communication network design, redundant radios on the infrastructure and vehicles, overlapping radio coverage and hot standby VCs; similar to the CBTC in the railways.
4. Emergency beacon solutions possibility / manual override from the vehicles, possibly extend for V2V communications.
5. Additional back – up communication system: e.g. add Vehicle to Vehicle (V2V) system to use the other vehicle as a relay to communicate to the infrastructure and based on the measured communications delay increase the headway;

The solution to the failure of the communication system is to perform the emergency brake. The emergency braking system is a key safety feature and needs to be designed such that vehicle can stop at any point at all speeds also in case of the communication systems failure. This does lead to a halt of operations.

In order to enable a redundant solution a self-sufficient solution a signaling system with a standardized set of signs could be investigated. These signs could be used by the vehicle in the event of such failure. This would allow that the vehicle in such event is able to maintain its trajectory with for example a slower speed and/or come to a safe stop within the trajectory.

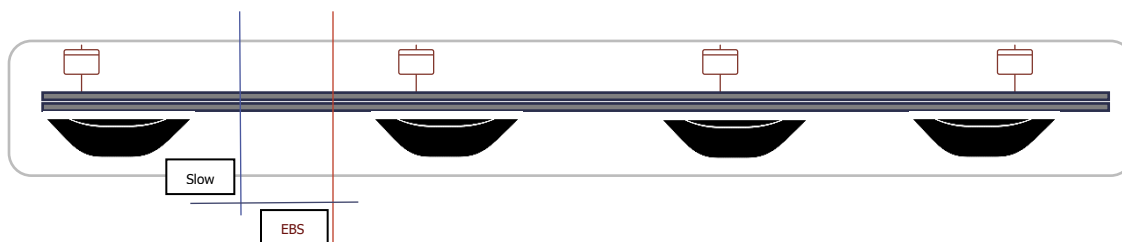


Figure 2. Example of the EBS and a potential signaling system to slow down the flows

The identified requirements are listed below:

| # | Description                                                                                                                                                                          | Impacted System | Impacted Sub-system |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------|
| 1 | The network-wide primary communication system shall not be expected to fail when considering the lifetime of a network                                                               |                 |                     |
| 2 | A section of the primary communication system shall not be expected to fail when considering the lifetime of a route, but might fail several times when considering the network      | Cargoloop: OCS  |                     |
| 3 | The primary communication system on a vehicle is not expected to fail when considering the lifetime of a set of vehicles, but might fail several times when considering all vehicles | Cargoloop: OCS  |                     |
| 4 | If the nominal communication system fails, the system shall ensure a safe halt of operation or safe (reduced) operations                                                             | OCS             |                     |
| 5 | The vehicles shall be able to detect emergency scenarios related to the vehicle ahead and trigger emergency procedures autonomously                                                  | OCS, VS         |                     |
| 6 | If a single, local fault/failure of the primary communication system occurs, the system shall allow continues operation                                                              | OCS, VS, TS     |                     |
| 7 | If the safe headway is not maintained, the vehicle shall be able to detect this and correct it autonomously                                                                          | OCS, VS, TS     |                     |

## Cargo Screening/Hub Security Payload related

This hazard group is related to the hazards related to the harmful payload or out of specification hazard.

The solutions is that the payload to be transported in cargoloop shall be screened prior to arrival at the cargobay.

Screening process and system shall ensure that the payload is packaged according to the standardized specifications and meets all the security requirements.

| # | Description                                                                                                                                   | Impacted System | Impacted Sub-system |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------|
| 1 | The payload shall be screened for packaging measures/ specifications and security                                                             |                 |                     |
| 2 | If the screened payload does not comply with the specification / security requirements it shall be rejected for entry to the cargoloop system |                 |                     |

## VIII. Recommendations

In addition to high-level safety requirements during the solution analysis process on the hazard topics several recommendations for future work are identified. These recommendations are listed below:

1. The baseline document needs to be updated into a new version namely 3.0 reflecting the findings during the developments of the safety concept;
2. The FHA needs to be revisited and reevaluated in the preliminary design phase;
3. Safety design levels need to be specified for hyperloop. The expectation is that the risk assessment method used during this concept phase and the well-known safety integrity levels (SIL) can be utilized as a starting point. Quantifying these levels will require a detailed design of the system as well as more in depth research on the relevant parameters. Preliminary design phase should therefore include specification of safety design levels;
4. Safe walk zones around the cargoloop system for the humans (personnel, visitors) as well as the safety during maintenance need to be further determined;
5. Removable cargobay is seen in terms of safety perspective a good solution for preventing hazards related to pressure interface failures;
6. Design a securing cargo system and / or identify a party that can develop a pallet slide / pallet stacking within the vehicle;
7. Communication Systems
  - a. Communications technology for hyperloop is essential for several key features of the system, not only for safety but also for the navigation, operations and maintenance. However, the applicability of the current radio/wireless communications systems is not a proven field in an hyperloop environment. As such it requires more study and research. Some hyperloop developers have published the first results<sup>3</sup> in this field. In the next phases the recommendation is to introduce a Communications Systems Infrastructure assignment unit.

Within this solutions space a layered approach could be considered. Starting from a simple solution and scaling up to a fully connected hyperloop. This would entail:

- OCS communicates to the guideway and there is a fixed predefined basic signaling to the vehicle, which the vehicle is able to identify;
- Coordination of the state of the vehicles through primary robust communications infrastructure between the OCS, guideway and vehicles
- Fully connected hyperloop with an extensive and robust communications infrastructure that allows for vehicle to vehicle communications;

The main goal should be to guarantee operation of the primary communication system. If the primary system should fail, a safe halt of operations should be guaranteed. To allow continuous network operability it is recommended to look into a solution of continuing operations in a reduced manner based on a secondary, simple system (e.g. hard coded signals, vehicle to vehicle sensing / autonomous operations).

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<sup>3</sup> <https://www.mdpi.com/1996-1073/14/4/983>

In addition it is necessary to look for a technology partner that will work within the WG Cargo / HDP in identifying the suitable technologies.

8. The cargo screening as well as hub security should not be part of the cargoloop system. The expectation is that the screening is done prior to the loading of payload in cargoloop.

- 9.

# IX. Appendix \_ Solution analysis template

Each of the hazard topics an aanalysis of different solutions have done following a template for solutions.

An example of such a template is inserted below:

|                    |                                                                                  |
|--------------------|----------------------------------------------------------------------------------|
| <b>Hazard Name</b> | Enter the name of the identified at system level hazard topic:                   |
|                    | Lack of protection against external threats and failure to control system access |

|                           |                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Description</b> | Describe the identified hazard system state:                                                                         |
|                           | Lack of protection against terrorist attacks in various forms initiating mainly by unauthorized access to the system |
|                           | Describe the end effect of the hazard from the FHA:                                                                  |
|                           | Loss of multiple vehicles and large sections of the guideway                                                         |

|                                     |                                         |      |         |           |      |                |  |  |  |
|-------------------------------------|-----------------------------------------|------|---------|-----------|------|----------------|--|--|--|
| <b>List Failure Mode Id</b>         | Enter only the FMID number from the FHA |      |         |           |      |                |  |  |  |
|                                     | FM39                                    | FM40 | FM41    | FM42      | FM43 | FM196          |  |  |  |
|                                     | Unauthorized access                     |      | Hacking | Explosion |      | Vehicle impact |  |  |  |
| <b>Severity</b>                     | 2                                       | 5    | 5       | 5         | 5    | 5              |  |  |  |
| <b>Likelihood</b>                   |                                         |      | 3       | 3         | 3    | 3              |  |  |  |
| <b>Safety Integrity Level (TBD)</b> |                                         |      |         |           |      |                |  |  |  |

|                          |                                                            |
|--------------------------|------------------------------------------------------------|
| <b>Associated Hazard</b> | Enter the name of the identified associated hazard:        |
|                          | Communication systems failure & Coordination of Operations |

| Solutions         | Enter all identified solutions for the hazard above:                                                                                                                                                                                                                                                                                                             |                    |                      |                        |          |                           |                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------|------------------------|----------|---------------------------|------------------------|
|                   | Description                                                                                                                                                                                                                                                                                                                                                      | Severity Reduction | Likelihood Reduction | Safety Integrity Level | Priority | Difficulty in realization | Cost                   |
| <b>Solution 1</b> | Only for FM39:<br><br>Having back-up/emergency access points with specific checks for personnel in place                                                                                                                                                                                                                                                         |                    |                      |                        |          |                           |                        |
| <b>Solution 2</b> | -Prevent unauthorized access to operational-related (restricted) areas<br><br>- Surveillance systems (possibly camera's) to monitor operational areas<br><br>- Prevent vehicles hitting the guideway (possibly ditches around support system)<br><br>-In case of unauthorized access, (parts of) the system will shut-down or go to emergency mode (lower speed) |                    |                      |                        |          |                           |                        |
| <b>Solution 3</b> | To avoid Hacking:<br><br>- Firewalls layers on communications networks<br><br>- Redundant communication networks<br><br>- Specific communications network only for the cargoloop system (hacking only possible close-by the system)                                                                                                                              |                    |                      |                        |          |                           |                        |
| <b>Solution 4</b> | - See "communication system failure" for reducing severity of a breach of communications                                                                                                                                                                                                                                                                         |                    |                      |                        |          |                           |                        |
| <b>Solution 5</b> | Explosion:<br><br>-Making the structure explosion-proof (Thicker structure or multiple layers or energy absorbing material)                                                                                                                                                                                                                                      |                    |                      |                        |          |                           | Very costly, adding to |

|  |  |  |  |  |  |  |                  |
|--|--|--|--|--|--|--|------------------|
|  |  |  |  |  |  |  | system footprint |
|--|--|--|--|--|--|--|------------------|

| Selected Solution | Enter the name of the selected solution and provide the reasoning behind the decision                                                                                                  |                       |                     |                          |                           |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------|--------------------------|---------------------------|
|                   |                                                                                                                                                                                        |                       |                     |                          |                           |
|                   | Description                                                                                                                                                                            | Impacted System (FBS) | Impacted Sub-system | Product Life-cycle phase | Verification & Validation |
| Requirement 1     | Unauthorized access to operational areas (against reasonably expected threats) shall be prevented                                                                                      |                       |                     |                          |                           |
| Requirement 3     | Vehicular access to the guideway by anyone (excl. Personnel) with vehicles commonly accessible to the public shall be prevented                                                        |                       |                     |                          |                           |
| Requirement 4     | Back-up access points shall be available for personnel to access the guideway in case of emergency                                                                                     |                       |                     |                          |                           |
| Requirement       | Safety areas for storage of stationary vehicles and maintenance tools as well as performing maintenance activities shall be envisioned at hubs as well as every x km along every route |                       |                     |                          |                           |
| Requirement 5     | Damages due to intentional harm shall not be anticipated to happen to a single route (<250km) in its entire lifespan                                                                   |                       |                     |                          |                           |
| Requirement 6     | In the event of unauthorized access                                                                                                                                                    |                       |                     |                          |                           |

|                      |                                                                                                                                                                                                    |  |  |  |  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|
|                      | (with risk of intentional harm) in and around the guideway, operations shall be halted on that route                                                                                               |  |  |  |  |
| <b>Requirement 6</b> | In the event of unauthorized access (with risk of intentional harm) in and around the guideway, operations on adjacent routes shall enter an emergency state (with lower speed and throughput TBD) |  |  |  |  |
| <b>Requirement 7</b> | The operational control system shall be designed to reduce the likelihood of hacking to less than x times during the operational lifetime of multiple vehicles                                     |  |  |  |  |
| <b>Requirement 8</b> | The Operational control system shall be designed to detect unauthorized cyber access                                                                                                               |  |  |  |  |
| <b>Requirement 9</b> | The Operational control system shall be designed to shutdown or halt operations in case unauthorized cyber access is detected                                                                      |  |  |  |  |

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